

Question 12.14

AI25BTECH11040 - Vivaan Parashar

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Question:

The eigenvalues of the matrix, $\mathbf{A}^{-1}$, where $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ are

a) 1 and $1/2$

c) 2 and 3

b) 1 and $1/3$

d) $1/2$ and $1/3$

Solution:

The eigenvalues of a matrix $\mathbf{A}$ are given by the roots of the characteristic polynomial, which is defined as:

$$\det(\mathbf{A} - \lambda \mathbf{I}) = 0$$

where λ represents the eigenvalues and $\mathbf{I}$ is the identity matrix of the same size as $\mathbf{A}$ ($\mathbf{I}_2$).

To find the eigenvalues of $\mathbf{A}^{-1}$, we can use the property that the eigenvalues of the inverse of a matrix are the reciprocals of the eigenvalues of the original matrix. This can be proven using the fact

$$\begin{aligned}\mathbf{A}\mathbf{v} &= \lambda\mathbf{v} \implies \mathbf{A}^{-1}\mathbf{A}\mathbf{v} = \mathbf{A}^{-1}(\lambda\mathbf{v}) \\ &\implies \mathbf{v} = \lambda\mathbf{A}^{-1}\mathbf{v} \\ &\implies \mathbf{A}^{-1}\mathbf{v} = \left(\frac{1}{\lambda}\right)\mathbf{v}\end{aligned}$$

Therefore, we first need to find the eigenvalues of $\mathbf{A}$. The characteristic polynomial of $\mathbf{A}$ is given by:

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \det \begin{vmatrix} 2 - \lambda & 1 \\ 0 & 3 - \lambda \end{vmatrix} = (2 - \lambda)(3 - \lambda) - 0 = (2 - \lambda)(3 - \lambda)$$

Setting this equal to zero gives us the eigenvalues of $\mathbf{A}$:

$$(2 - \lambda)(3 - \lambda) = 0$$

Thus, the eigenvalues of $\mathbf{A}$ are $\lambda_1 = 2$ and $\lambda_2 = 3$.

Now, the eigenvalues of $\mathbf{A}^{-1}$ are the reciprocals of these eigenvalues:

$$\mu_1 = \frac{1}{\lambda_1} = \frac{1}{2}, \quad \mu_2 = \frac{1}{\lambda_2} = \frac{1}{3}$$

Therefore, the eigenvalues of $\mathbf{A}^{-1}$ are $\frac{1}{2}$ and $\frac{1}{3}$. The correct answer is option (d): $1/2$ and $1/3$.